

Chapter 7

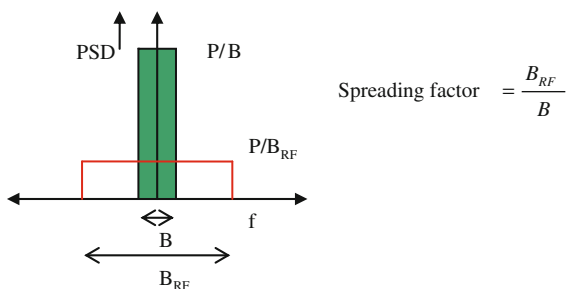
Spread Spectrum Modulation

7.1 Introduction

In the technique of spread spectrum, the frequency spectrum of a data-signal (shaded in Fig. 7.1) is spread (non-shaded) using a code uncorrelated with that signal. As a result the bandwidth occupancy is much higher than required [4].

The codes used for spreading have low cross-correlation values and are unique to every user. This is the reason that a receiver which has knowledge about the code of the intended transmitter is capable of selecting the desired signal (Fig. 7.1).

Fig. 7.1 PSD plot before and after spreading the signal



The points to be noted here are as follows:-

- ✓ The spread (i.e., modulated) signal can be de-spread by the intended receiver only.
- ✓ Spread spectrum technique is actually 2nd time modulation over an already modulated signal.

Say, an AM signal of power P_S has bandwidth 10 kHz. After spreading, it will have bandwidth 1 MHz.

Therefore, after spreading, the power is

$$P_{sp} = \frac{P_S}{\frac{10^6}{10^4}} = \frac{P_S}{100}$$

In dB,

$$\left. \frac{P_S}{100} \right\|_{dB} = 10 \log_{10} P_S - 10 \log_{10} 100 = P_S \|_{dB} - 20 \text{ dB}$$

∴ Spread spectrum (modulated twice) provides noise 20 dB lower than the power of AM (modulated once).

A number of advantages are [2]:

- *Low power spectral density.* As the signal is spread over a large frequency-band, the Power Spectral Density is getting very small, so other communications systems do not suffer from this kind of communications. However the Gaussian Noise level is increasing.
- *Interference limited operation.* In all situations the whole frequency-spectrum is used.
- *Privacy due to unknown random codes.* The applied codes are - in principle – unknown to a hostile user. This means that it is hardly possible to detect the message of another user.
- *Applying spread spectrum implies the reduction of multi-path effects.*
- *Random access possibilities.* Users can start their transmission at any arbitrary time.
- Good anti-jam performance.

Because of the difficulty to jam or detect spread spectrum signals, the first applications were in the military field. However nowadays spread spectrum systems are gaining popularity also in commercial applications.

7.2 Processing Gain

The main parameter in spread spectrum systems is the processing gain: the ratio of transmission and information bandwidth:

$$G_p = \frac{B_{RF}}{B} \quad (7.1)$$

which is basically the ‘spreading factor’. The processing gain determines the number of users that can be allowed in a system, the amount of multi-path effect reduction, the difficulty to jam or detect a signal etc. For spread spectrum systems it is advantageous to have a processing gain as high as possible.

There exist different techniques to spread a signal:

1. Direct-Sequence (DS),
2. Frequency-Hopping (FH),
3. Time-Hopping (TH) and
4. Hybrid spread spectrum

We will concentrate on the two most popular techniques: Direct-Sequence and Frequency-Hopping, in this chapter.

7.3 Pseudo-Noise (PN) Sequence

PN sequence is the code used for spreading the signal of transmission. The frequency of the sequence is much higher with that of the transmitting signal. Actually the PN sequence is noise like, and periodic over one window. To understand the concept of pseudo-noise sequence let's consider one hypothetical experiment.

7.3.1 Concept: A Hypothetical Experiment

Let us consider a case where a faulty diode is generating random and high frequency noise with zero mean. For particular time duration, I have taken the noise in a form of a window. Now, the windows are concatenated side by side, one by one. These concatenated windows are given to one of the inputs of a zero-crossing detector (Fig. 7.2).

Thus, at the output, we will get a high frequency rectangular signal. Now, if we name the smallest chip duration (smallest high value time or smallest low value time) as logic 1 or logic 0 respectively, then we can get one chip sequence as 0011101| 0011101 | 001110...and so on (as the same windowed noise is concatenated, the sequence also made periodic).

From the sequence it's clear, that, the sequence is noise like random within a window, but periodic. That's why the sequence can be called as pseudo-random sequence (pseudo means 'not real').

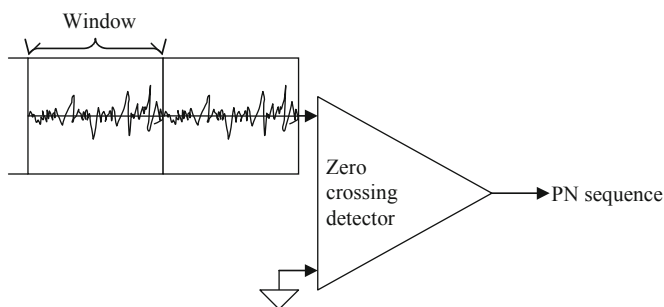
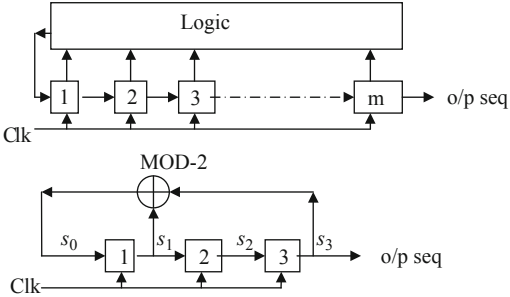


Fig. 7.2 Hypothetical experiment on concept of PN sequence

7.3.2 Generation of PN Sequence

In Fig. 7.3a, an m-bit length serial in parallel out shift register is shown [2, 5]. From all the flip-flops of the shift register, outputs are fed to one logic circuit with one switch to each. The output of the logic circuit is again fed to the input of the primary (left most) flip-flop. This arrangement is called as *feedback shift register* arrangement.

Fig. 7.3 (a) Feed back shift Register. (b) PN sequence generator



PN sequence can be easily generated by using the feedback register arrangement. Here I have considered 3 flip-flops with outputs s_1, s_2, s_3 . Therefore the length of the shift register is $m = 3$. The output of the entire system is s_3 . Its also can be said from Fig. 7.3b that, after a clock pulse is encountered, $s_2 = s_1, s_3 = s_2, s_1 = s_1 \oplus s_3$. The entire system state table is as follows, taking initial contents of the shift register ‘1111’:

From Table 7.1, the shift register configuration shown in the Fig. 7.3b, is generating a sequence ‘1110100’ followed by the same sequence again and again,

Table 7.1 State table for PN sequence generation

Clock pulse no.	Intermediate states			Output
	S ₁	S ₂	S ₃	
1	1	1	1	1
2	0	1	1	1
3	1	0	1	1
4	0	1	0	0
5	0	0	1	1
6	1	0	0	0
7	1	1	0	0
8	1	1	1	1

periodically. It is observed that after 8th clock pulse we are getting the same initialized flip flop content '111'.

Note: MOD-2 addition:

MOD-N means periodic with periodicity N. Therefore MOD-2 operation generates a window of sequence of length 2. To represent 2 patterns we need only one bit (0 and 1). Let's consider MOD-2 addition (i.e., one bit number + one bit number = one bit number).

$$\begin{aligned} 0 + 0 &= 0 \\ 0 + 1 &= 1 \\ 1 + 0 &= 1 \\ 1 + 1 &= 10 \end{aligned}$$

The last result (1+1) is generating two bit result. The MSB is called as CARRY and the LSB is the SUM. To follow the procedural principle of MOD-2 addition, we must have only 1 bit. Therefore, by ignoring the CARRY bit, the result of (1+1) operation is made equal to 0. The mod-2 addition is symbolized by $\oplus$. The truth table is shown in Table 7.2.

Table 7.2 Table of MOD-2 operation is following the XOR gate truth table

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

From the Table 7.2, it is very clear that, MOD-2 operation is following the XOR gate truth table. Therefore the MOD-2 operation is realized practically by XOR gate.

7.3.3 Properties of PN Sequence

1. In each period, Number of 1s = Number of zeroes +1;
2. If the number of flip-flop is m, the sequence length N must be, $N = (2^m - 1)$
3. At any instant, the square of the PN sequence $c(t)$ is 1. $c^2(t) = 1$.
4. Autocorrelation

$$R_c(\tau) = \frac{1}{T_c} \int_{-\frac{T_c}{2}}^{\frac{T_c}{2}} c(t)c(t-\tau)dt,$$

where T_c is the chip duration of the PN sequence. $T_c \ll T_b$ (bit duration). The derivation of auto correlation can be derived from the table as follows:

Table 7.3 Autocorrelation function table of PN sequence

Shift	Original/shifted	Σ	$R_c(\tau) = \Sigma / N$
0	1110100 1110100	$1+1+1+1+1+1=7$	$7/7=1$
1	1110100 0111010	$-1+1+1-1-1-1+1=-1$	$-1/7=-0.14$
2	1110100 0011101	$-1-1+1-1+1-1+1=-1$	$-1/7=-0.14$
3	1110100 1001110	$+1-1-1-1+1-1+1=-1$	$-1/7=-0.14$
4	1110100 0100111	$-1+1-1+1+1-1-1=-1$	$-1/7=-0.14$
5	1110100 1010011	$+1-1+1+1-1-1-1=-1$	$-1/7=-0.14$
6	1110100 1101001	$+1+1-1-1-1+1-1=-1$	$-1/7=-0.14$
7	1110100 1110111	$+1+1+1+1+1=7$	$7/7=1$
8	1110100 0111010	$-1+1+1-1-1-1+1=-1$	$-1/7=-0.14$

The graph obtained from the Table 7.3, is as follows (Fig. 7.4).

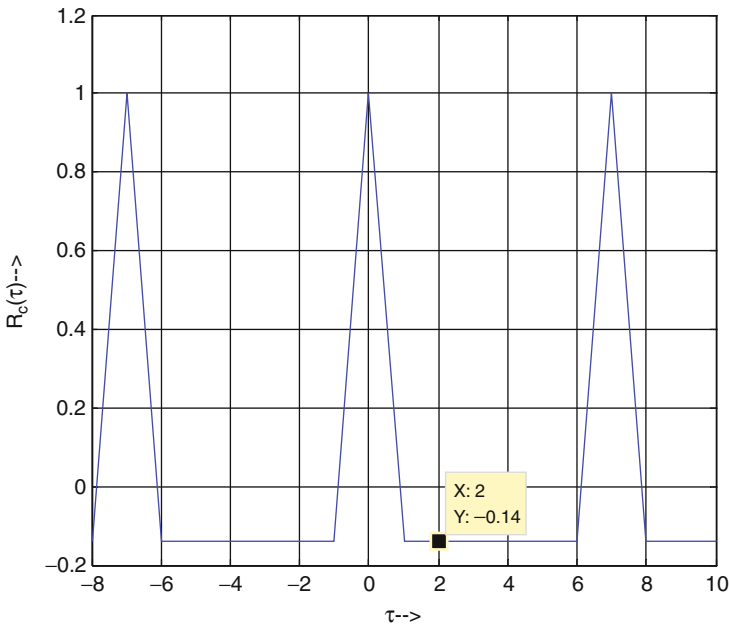


Fig. 7.4 Autocorrelation of PN sequence

7.4 Direct Sequence Spread Spectrum (DSSS)

7.4.1 Concept

Direct Sequence is the best known Spread Spectrum Technique [4]. Here, the data signal (with bit duration T_B) is multiplied by a Pseudo Random Noise Code (PN code) with chip duration T_C , where $T_C \ll T_B$.

A PN code is a sequence of chips valued -1 and 1 (polar) or 0 and 1 (non-polar) and has noise-like properties. This results in low cross-correlation values among the codes and the difficulty to jam or detect a data message.

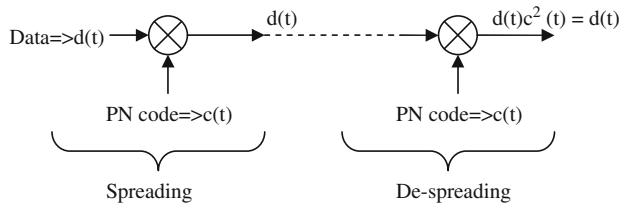


Fig. 7.5 Direct sequence spread spectrum

As shown in the Fig. 7.5, the data $d(t)$ is multiplied by the high frequency chip sequence (PN code) $c(t)$. It results in bandwidth spreading.

$$\text{As, } c^2(t) = 1,$$

if, the spread sequence is again multiplied by $c(t)$, the output of the receiver would be $d(t)c^2(t) = d(t)$. The chips are actually chopping the bit sequence s shown in the Fig. 7.6.

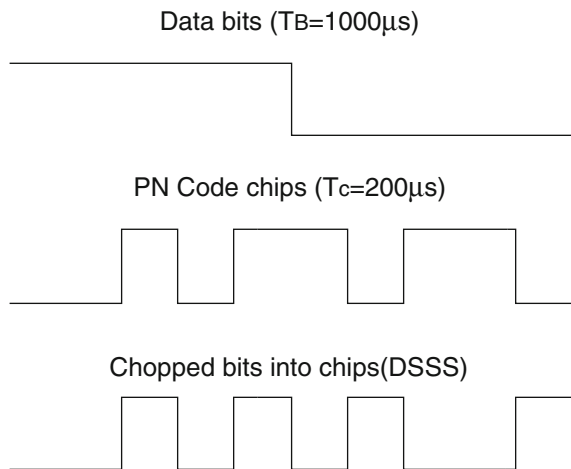


Fig. 7.6 Direct-sequence spreading by bit chopping

As the bandwidth of the data signal is now multiplied by a factor G_p . The power contents however stays the same, with the result that the power spectral density lowers.

Here, the spreading is done by multiplying the digital data with PN chip sequence. This is very much analogous to Amplitude Modulation technique as (1) the modulation is done by simple multiplication and (2) the amplitude of the data is being modified/modulated, both of which are characteristics of Amplitude Modulation (AM).

For this reason it is easy to introduce a large processing-gain in Direct-Sequence systems.

In the receiver, the received signal is multiplied again by the same (synchronized) PN code. Since the code existed of +1s and -1s, this operation completely removes the code from the signal and the original data-signal is left. Another observation is that the de-spread operation is the same as the spread operation. The consequence is that a possible jamming-signal in the radio channel will be spread before data-detection is performed. So jamming effect is reduced (see Fig. 7.7).

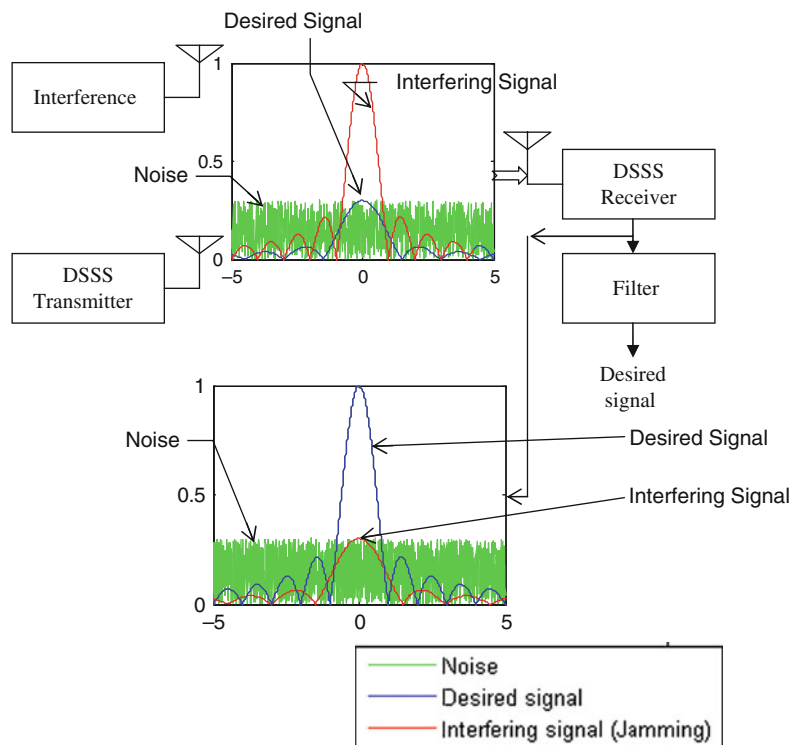


Fig. 7.7 DS-concept, before and after despreading

The main problem with applying Direct Sequence spreading is the so-called Near-Far effect which is illustrated in Fig. 7.8. This effect is present when an

interfering transmitter is much closer to the receiver than the intended transmitter. Although the cross-correlation between codes A and B is low, the correlation between the received signal from the interfering transmitter and code A can be higher than the correlation between the received signal from the intended transmitter and code A. The result is that proper data detection is not possible.

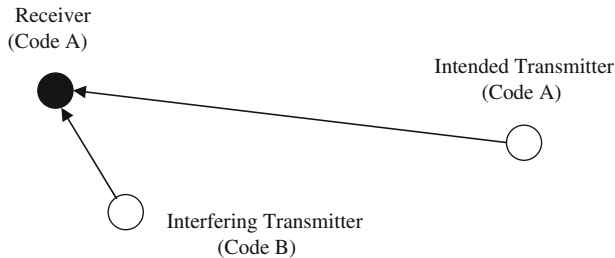


Fig. 7.8 Illustration of Near-far effect

7.4.2 DSSS with Coherent BPSK

As we know from previous chapters, Binary Phase Shift Keying is a very important digital modulation technique in practice. Here one BPSK modulate signal is treated as the information $s(t)$ and multiplied by the chip sequence (PN code) $c(t)$. As DSSS is also a modulation technique, we can infer, this technique of spread spectrum modulation is actually 2nd time modulation over an already modulated wave.

From the Fig. 7.9, we get the BPSK signal $s(t)$. The signal transmitted through the channel is $s(t)c(t)$. Additive white Gaussian Noise (AWGN) $n(t)$ is added in the channel. Therefore the received signal is

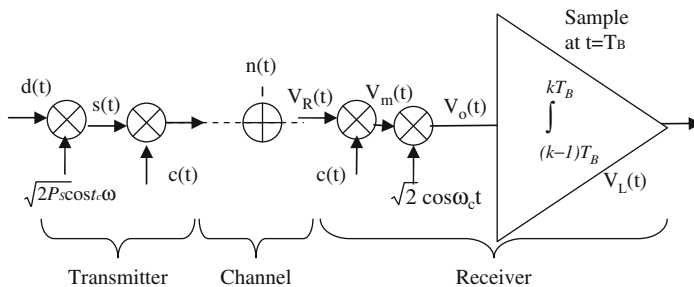


Fig. 7.9 BPSK incorporating DSSS

$$\begin{aligned}
v_R(t) &= s(t)c(t) + n(t) \\
\Rightarrow v_R(t) &= \sqrt{2P_S}d(t)c(t) \cos \omega_c t + n(t)
\end{aligned} \tag{7.2}$$

After multiplying the received signal by PN chip code $c(t)$ we get,

$$\begin{aligned}
v_m(t) &= \sqrt{2P_S}d(t)c^2(t) \cos \omega_c t + n(t)c(t) \\
\Rightarrow v_m(t) &= \sqrt{2P_S}d(t) \cos \omega_c t + n(t)c(t)
\end{aligned} \tag{7.3}$$

The 2nd term of the Eq. (7.3) indicates the noise is chopped, i.e. (1) noise is spread whereas wanted data is de-spread and hence (2) power spectral density lowers as shown in the Fig. 7.7.

Next, $v_m(t)$ is again passed through another multiplier, integrator, i.e. BPSK detector. The probed signals are as follows.

$$v_o(t) = \sqrt{P_S}(2 \cos^2 \omega_c t)d(t) + n(t)c(t) \cos \omega_c t \tag{7.4}$$

The signal $v_o(t)$ is then sampled at $t=kT_B$ to get the final output $v_L(t)$.

$$\begin{aligned}
v_L(t) &= \int_0^{T_B} v_o(t)dt; k = 1 \\
\Rightarrow v_L(t) &= \int_0^{T_B} \left[\sqrt{P_S}(2 \cos^2 \omega_c t)d(t) + n(t)c(t) \cos \omega_c t \right] dt = d(t)
\end{aligned} \tag{7.5}$$

By this way the data signal $d(t)$ has been correctly detected at the end output.

7.4.3 Probability of Error Calculation

From the discussions of Chaps 3 and 6, it has been shown that, the probability of error of BPSK for matched filter reception without employing spread spectrum is

$$P_e = \frac{1}{2} \operatorname{erfc} \left[\frac{E_S}{\eta} \right]^{1/2} \tag{7.6}$$

where, E_S is the symbol energy, $\eta/2$ is the two sided power spectral density (variance) of the noise.

To consider the case of direct sequence spread spectrum incorporated with BPSK (Eq. 7.2), within bit interval $0 \leq t \leq T_B$, the transmitted signal is

$$s_t(t) = d(t)g(t)c(t) \cos 2\pi f_c t; 0 \leq t \leq T_B \tag{7.7}$$

where, $d(t)$ is the information symbol and the pulse $g(t)$ is defined from the Eq. (7.2) as

$$g(t) = \begin{cases} \sqrt{2P_S} = \sqrt{\frac{2E_S}{T_B}}; & 0 \leq t \leq T_B \\ 0; & \text{otherwise} \end{cases} \quad (7.8)$$

and $c(t)$ is the PN code generated, in the bit interval $0 \leq t \leq T_B$ expressed as

$$c(t) = \sum_{n=0}^{L_C-1} c_n p(t - nT_C) \quad (7.9)$$

Here,

L_C = number of chips per bit

$\{c_n\}$ = PN sequence

The code chip sequence is uncorrelated (white), i.e.,

$$E(c_n c_m) = 0; \quad \text{for } n \neq m \quad (7.10)$$

and each chip is +1 or -1 with probability 0.5.

$$\therefore E(c_n) = 0, \quad \text{and} \quad E(c_n^2) = 1 \quad (7.11)$$

When the signal is interfered by an additive signal $n(t)$, the received signal is

$$r(t) = d(t)g(t)c(t) \cos 2\pi f_c t + n(t) \quad (7.12)$$

Since the received signal is the output of a bandpass signal, $n(t)$ must have only the band pass components, i.e.,

$$n(t) = n_c(t) \cos 2\pi f_c t - n_s(t) \sin 2\pi f_c t \quad (7.13)$$

Equation (7.13) is the expression of bandpass random noise. $n_c(t)$ and $n_s(t)$ are the two quadrature components.

After this, the received signal is multiplied by $c(t)$ and goes through an integrator, as shown in Fig. 7.9. Taking out only the received noise part from Eq. (7.4), sampled at $t = T_B$,

$$\begin{aligned}
y_n(T_B) &= \int_0^{T_B} c(t)n(t)g(t) \cos(2\pi f_c t) dt \\
&= \sum_{n=0}^{L_C-1} c_n \int_0^{T_B} p(t - nT_c)n(t)g(t) \cos(2\pi f_c t) dt \\
&= \sqrt{\frac{2E_S}{T_B}} \sum_{n=0}^{L_C-1} c_n s_n
\end{aligned} \tag{7.14}$$

where,

$$s_n = \int_{nT_c}^{(n+1)T_c} n(t) \cos(2\pi f_c t) dt \tag{7.15}$$

The probability of error directly depends upon the statistical parameters of the interfering signal. The mean of the noise is

$$E[y_n(T_B)] = 0 \tag{7.16}$$

And the variance is

$$E[y_n^2(T_B)] = \frac{2E_S}{T_B} \sum_{n=0}^{L_C-1} \sum_{m=0}^{L_C-1} E(c_n c_m) E(s_n s_m)$$

As $E(c_n c_m) = \delta_{nm}$, as the property of PN sequence suggests, therefore,

$$E[y_n^2(T_B)] = \frac{2E_S}{T_B} \sum_{n=0}^{L_C-1} E(s_n^2) \tag{7.17}$$

Now, considering a single tone interference, i.e.,

$$n(t) = \sqrt{2P_N} \cos 2\pi f_c t \tag{7.18}$$

where, P_N is the average noise power. Substituting Eq. (7.18) into Eq. (7.15),

$$s_n = \sqrt{2P_N} \int_{nT_c}^{(n+1)T_c} \cos^2(2\pi f_c t) dt = \frac{\sqrt{2P_N}}{2} \int_{nT_c}^{(n+1)T_c} dt = \frac{T_c \sqrt{P_N}}{\sqrt{2}} \tag{7.19}$$

Here also, mean of noise

$$E(s_n) = E\left(\frac{T_c \sqrt{P_N}}{\sqrt{2}}\right) = 0 \quad (7.20)$$

The mean square value of noise is

$$E(s_n^2) = \frac{T_c^2 P_N}{2} \quad (7.21)$$

From Eqs. (7.21) and (7.17), we get the noise power/variance is

$$E[y_n^2(T_B)] = \frac{E_S P_N T_c}{2} \quad (7.22)$$

The ratio of signal power and noise power is the signal-to-noise ratio at the detector expressed as

$$\text{SNR} = \frac{E_S^2}{E_S P_N T_c / 2} = \frac{2E_S}{P_N T_c} \quad (7.23)$$

If the noise variance $\eta/2$ of Eq. (7.6) is same as $\frac{E_S P_N T_c}{2}$, then the probability of error of the system is

$$P_e = \frac{1}{2} \operatorname{erfc}\left(\frac{E_S}{E_S P_N T_c}\right)^{1/2} = \frac{1}{2} \operatorname{erfc}\left(\frac{f_c}{P_N}\right)^{1/2} \quad (7.24)$$

The expression of probability of error signifies, as chip-frequency is very high, the probability of error must be very low, according to the nature of the *erfc* graph shown in the Chap. 3.

7.5 Frequency-Hopped Spread Spectrum

7.5.1 Concept

Remembering FM in analog communication or FSK in digital communication, the concept of frequency hopped spread spectrum (*FHSS*) can be understood. In FM, the instantaneous carrier frequency is changed according to the baseband signal, whereas in FHSS, the input carrier frequency to the modulator itself varies within a fixed bandwidth. With respect to time, the frequency assigned for modulation is changed with a central frequency but with a fixed bandwidth. As the frequencies are changed from one to another and the allocation of frequency is pseudorandom, i.e., not in order (dependent upon a PN sequence), the term ‘*hopping*’ comes to represent the allocation of frequency with respect to time [6]. By this carrier frequency hopping, obviously the bandwidth of the signal increased.

The process of frequency hopping is illustrated in Fig. 7.10. A disadvantage of Frequency-Hopping over Direct-Sequence is that obtaining a high processing-gain is hard. The faster the ‘hopping-rate’ is, the higher the processing gain.

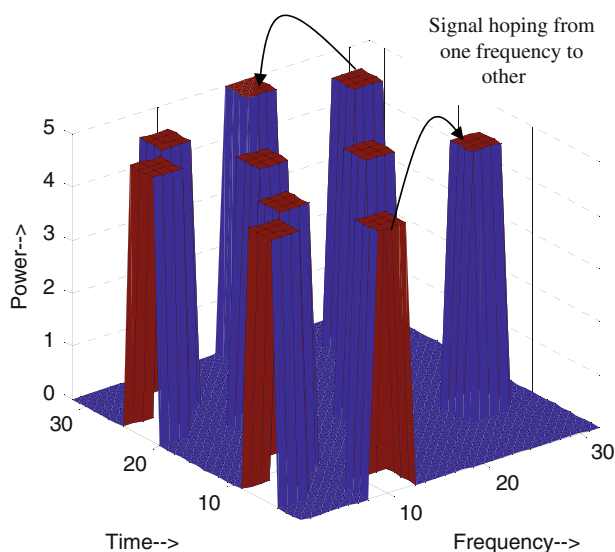


Fig. 7.10 Illustration of the frequency hopping concept

On the other hand, Frequency-Hopping is less affected by the Near-Far effect than Direct-Sequence. Frequency-Hopping sequences have only a limited number of ‘hits’ with each other. This means that if a near-interferer is present, only a number of ‘frequency-hops’ will be blocked instead of the whole signal. From the ‘hops’ that are not blocked it should be possible to recover the original data-message.

Figure 7.11 shows the general scheme of FHSS modulation. A PN sequence generator creates a k -bit pattern for every hopping period, T_h . The frequency table uses the pattern to find the frequency to be used for a particular hopping period and passes it to the frequency synthesizer. The frequency synthesizer creates a carrier signal of that frequency, and the source signal modulates the carrier signal.

Let’s consider a case (for theoretical illustration) where the PN code generator creates three bit patterns (8 different patterns). The 8 patterns generated by PN code generator are mapped to eight different frequencies in the frequency table (Fig. 7.12).

The pattern for this station is 101, 111, 001, 000, 010, 011, and 100. Note that the pattern is pseudorandom it is repeated after eight hoppings. This means that at hopping period 1, the pattern is 101. The frequency selected is 700 kHz; the source signal modulates this carrier frequency [3]. The second k -bit pattern selected

Fig. 7.11 FHSS modulator

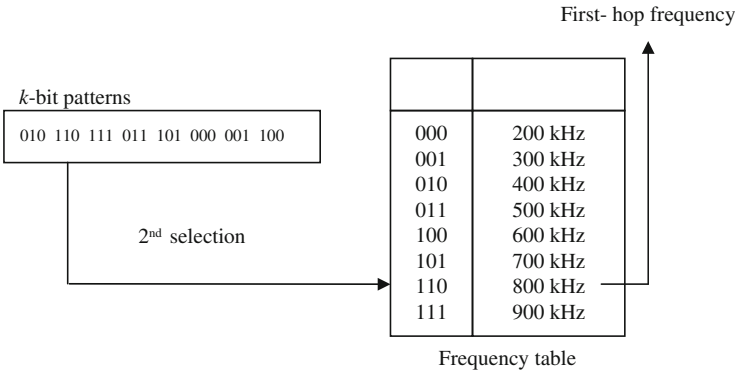
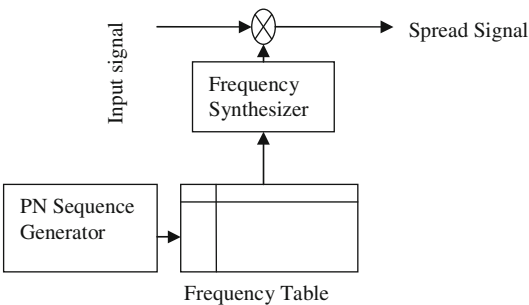


Fig. 7.12 Frequency selection in FHSS

is 111, which selects the 900-kHz carrier; the eighth pattern is 100, the frequency is 600 kHz. After eight hoppings, the pattern repeats, starting from 101 again.¹

7.5.2 FHSS with FSK

In short, the technique of integrated modulation by FHSS and FSK is discussed with proper block diagram representation of transmitter and receiver. As we know from the chapter of digital modulation, FSK is very much like FM. The carrier frequency used for frequency modulation is here dynamically determined by the PN sequence generator as discussed in the previous section.

We know that, the general FSK (before applying FHSS) can be expressed as

$$s_{\text{FSK}}(t) = \sqrt{2P_S} \cos \{ \omega_c t + d(t) \times \Omega t + \theta \} \tag{7.25}$$

¹Coherent detection of FHSS is possible only within each hop. Therefore, non-coherent detection is employed.

After the second modulation by FHSS of the already modulated (FSK) signal, the signal becomes:

$$s_{FSK_FHSS}(t) = \sqrt{2P_S} \cos \{\omega_i t + d(t) \times \Omega t + \theta\} \quad (7.26)$$

where, ω_i =instantaneous carrier frequency. The resultant signal is therefore responsible for *instantaneous transmission bandwidth*.

As shown in Fig. 7.13, here also, successive k bit segments of PN sequence drive the frequency synthesizer which enables the carrier frequency to hop over 2^k distinct values.

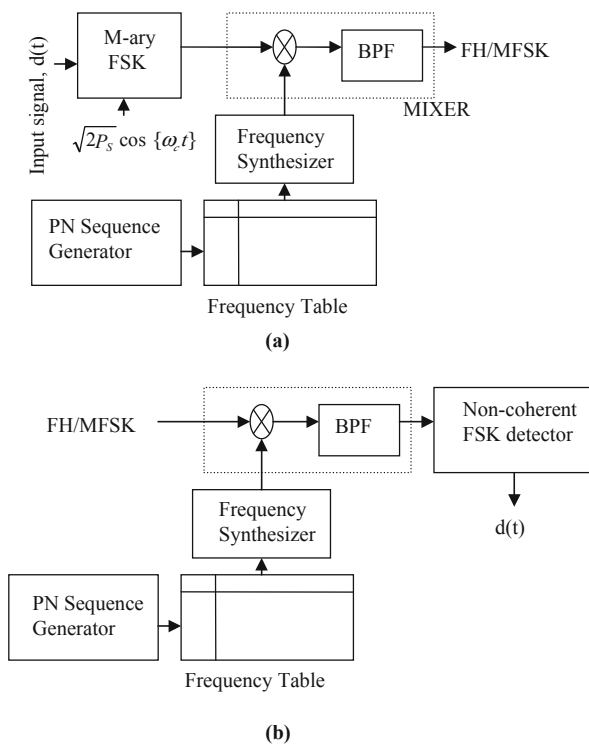


Fig. 7.13 Frequency hop spread M-ary FSK. (a) Transmitter, (b) Receiver

Each hop is obviously of conventional M-FSK. Therefore, through complete range of 2^k hops, the MFSK signal occupies much higher bandwidth.

We know from the previous discussion that, coherent detection is possible only within each hop, as the pilot carrier frequency is being changed according to the generated PN sequence. Therefore, non-coherent detection is employed.

7.5.3 Rate of Hopping: Fast and Slow

In FHSS, the available channel bandwidth W is sub-divided into a large number of non-overlapping (orthogonal) frequency slots. In any signaling interval, the transmitted signal occupies one or more of the available frequency slots. The selection of frequency slots in each signaling interval is made pseudorandomly according to the output from the PN generator. This pseudo non-deterministic change or jump of frequency is *frequency hopping* and the rate of frequency slot change is the hopping rate (R_h). As for an example in the case of binary FSK ($M = 2$), the modulator selects 2 frequencies corresponding to transmission of '0' or a '1' symbol [6].

The term *slow frequency hopping (SFH)* means there are several modulation symbols per hop. Similarly, *fast frequency hopping (FFH)* means there are several frequency hops per modulation symbol. For SFH, the shortest uninterrupted waveform of the system is that of the data symbol; whereas, for FFH, the shortest uninterrupted waveform of the system is that of the hop.

As an example of FFH, the aforementioned binary FSK system is taken. According to the Fig. 7.14a, there are 4 chips transmitted per bit. Here, for FFH, the chip duration itself is the hop duration. Fig. 7.14b illustrates an example of SFH binary FSK system. In this case, there are 3 bits transmitted during the time duration of single hop. Here, the hop duration is three times the chip duration. If we employ a 4-ary FSK ($M = 4$), then the number of bits/MSK symbol is $k = 2$. If 3 PN sequence chips are accommodated into single hop, then frequency allocation diagram shown in Fig. 7.15 can be obtained. In this case, the symbol duration (duration of 4 bits) and the hop duration will be identical.

7.6 Application of Spread Spectrum

7.6.1 GPS (Global Positioning System)

The global positioning system (GPS) uses direct sequence spread spectrum and operates on the principle of one way *ranging*. It is divided into three segments namely, space, control and user segment. The selected satellite orbital constellation contains 24 satellites in 6 orbital planes. The six orbital planes are 60° apart from each other containing 4 satellites each. The satellites have a period of 12 h sidereal time and a semi-major axis of 26561.75 km. One sidereal day means 23 h 56 min 4.09s which is the interval between two successive transits of the vernal equinox and is defined as the mean sidereal day. The four satellites are unevenly placed in each plane so as to minimize the effect of satellite outage. The eccentricity of the orbit is nearly zero and the inclination of the orbital plane is 55° . This satellite constellation allows at least four satellites to be visible to a user anywhere anytime on the earth. Table 7.4 lists the orbital parameters of the GPS constellation. The satellite visibility is also depicted in Fig. 7.16.

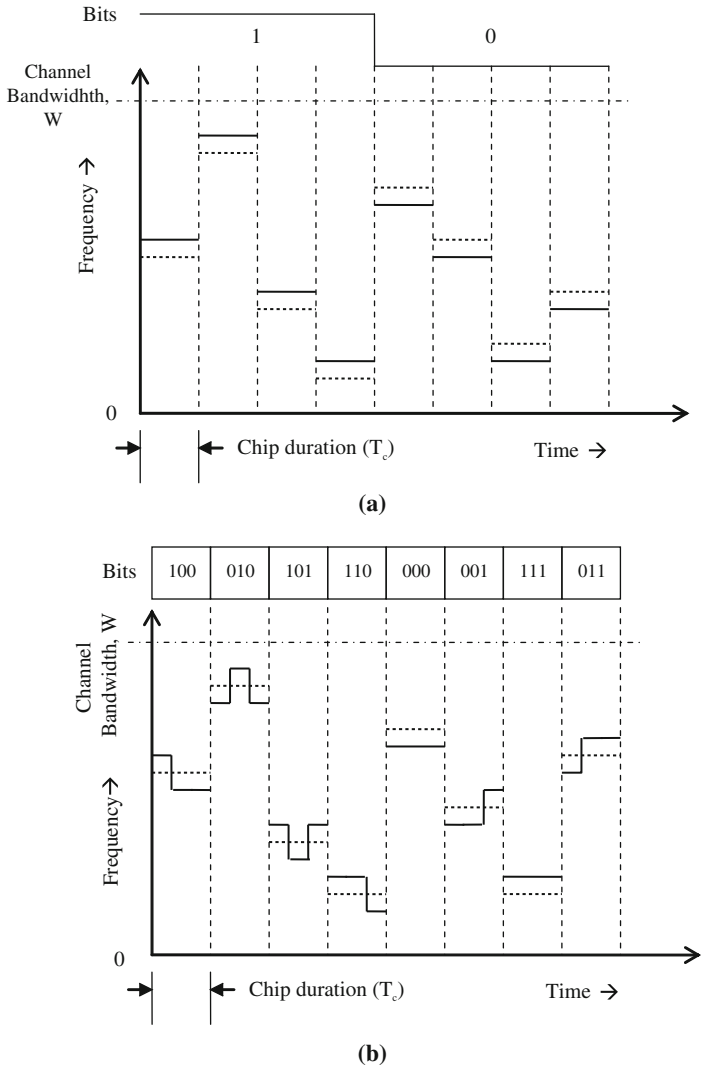


Fig. 7.14 Fast hopping versus slow hopping in FHSS system: (a) Fast hopping example: 4 hops/bit; (b) Slow hopping example: 3 bits/hop

GPS transmits at two frequencies in the L band, L1 at centre frequency 1,575.42 MHz and L2 at centre frequency 1,227.6 MHz (each being a multiple of 10.23 MHz). The L1 signal is bi-phase modulated. The spreading code with I channel is a Gold code of length 1,023 chips. The rate is 1.023 MHz and code period is 1 ms. Each satellite uses pre-assigned gold code. The code is called as course/acquisition or simply C/A code. The Q channel spreading code is a long non-linear code frequency 10.23 MHz. The code is called as precision or P code.

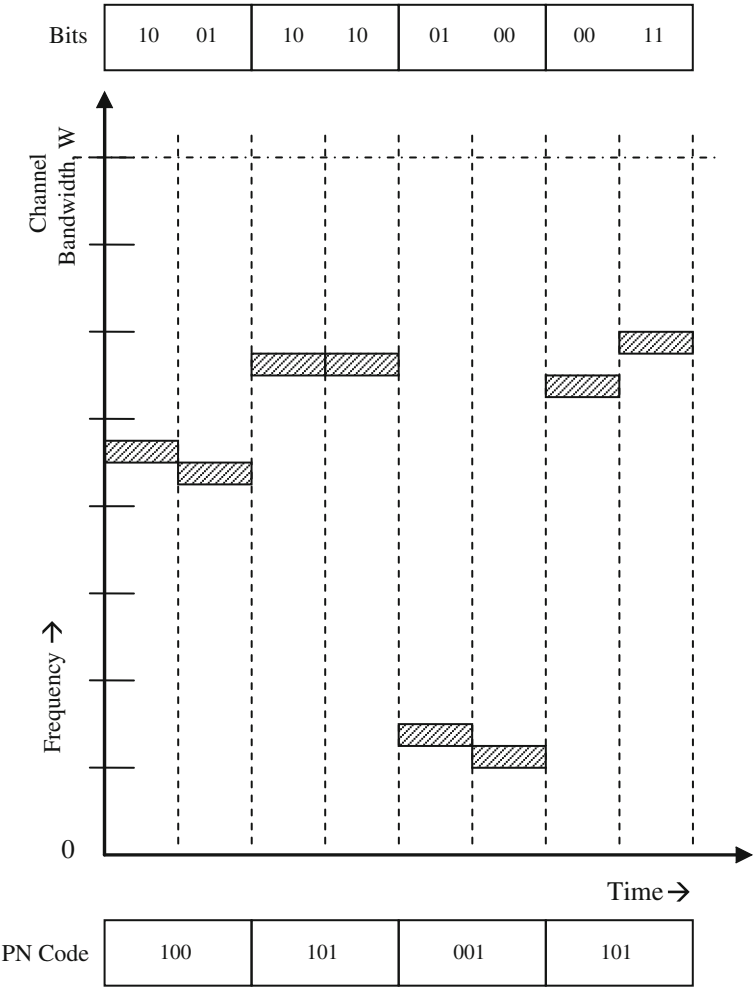


Fig. 7.15 Frequency variation over a complete period of the PN sequence

It is transmitted at a rate of 50 bps and repeated cyclically. The message contains information about the positions of the satellites, data to correct errors in propagation time, satellite clock bias, etc. a method of producing L1 signal is depicted in the Fig. 7.17.

The control segment consists of 5 monitor stations (Hawaii, Colorado Springs, Ascension Island, Diego Gracia and Kwajalan Island), four ground stations and operational control centre. All the monitor stations except Hawaii contain ground antenna upload stations. The clocks of the satellites are synchronized by a ground station of California. It also supplies the positions of the satellites obtained from

Table 7.4. Orbital parameters of the GPS constellation

No.	Satellite identifi- cation	Semi- major axis	Inclination, θ	Right ascension of ascending node	Mean anomaly	Phase difference	Longitude of ascending node
1.	A3	26,561.75	55.0	272.847	11.676	103.55	179.63
2.	A4	26,561.75	55.0	272.847	41.806	31.13	14.69
3.	A2	26,561.75	55.0	272.847	161.786	119.98	74.68
4.	A1	26,561.75	55.0	272.847	268.126	106.34	127.85
5.	B1	26,561.75	55.0	272.847	80.956	130.98	94.27
6.	B2	26,561.75	55.0	272.847	173.336	92.38	140.46
7.	B4	26,561.75	55.0	272.847	204.376	31.04	155.98
8.	B3	26,561.75	55.0	272.847	309.976	105.06	28.78
9.	C1	26,561.75	55.0	272.847	111.876	100.08	169.73
10.	C4	26,561.75	55.0	272.847	241.556	129.68	54.57
11.	C3	26,561.75	55.0	272.847	339.666	98.11	103.62
12.	C2	26,561.75	55.0	272.847	11.796	32.13	119.69
13.	D1	26,561.75	55.0	272.847	135.226	100.07	61.40
14.	D4	26,561.75	55.0	272.847	167.356	32.13	77.47
15.	D2	26,561.75	55.0	272.847	265.446	98.09	127.51
16.	D3	26,561.75	55.0	272.847	35.156	129.71	11.37
17.	E1	26,561.75	55.0	272.847	197.046	130.98	152.31
18.	E2	26,561.75	55.0	272.847	302.596	105.55	25.09
19.	E4	26,561.75	55.0	272.847	333.686	31.09	40.63
20.	E3	26,561.75	55.0	272.847	66.066	92.38	6.82
21.	F1	26,561.75	55.0	272.847	238.886	103.54	53.23
22.	F2	26,561.75	55.0	272.847	345.226	106.34	106.40
23.	F3	26,561.75	55.0	272.847	105.206	119.98	166.39
24.	F4	26,561.75	55.0	272.847	135.346	30.00	1.46

tracker system. Therefore, each satellite knows the position of all GPS satellites at any instant.

The user segment consists of GPS receiver. The receiver consists of a pre-amplifier, antenna, down converter, A/D converter, IF amplifier, navigation processing unit.



Fig. 7.16 GPS constellation in motion with the rotation of the earth

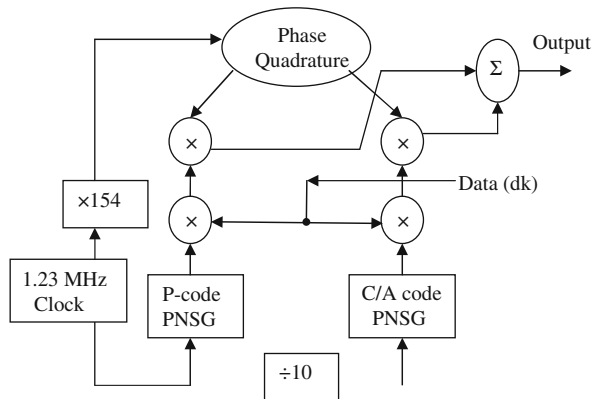


Fig. 7.17 Arrangement of producing L1 signal

Circular or rectangular microstrip is the common choice of antenna. The received GPS signal is very weak (of the order of -130 dB). But the reception is possible as the characteristics is spread spectrum, even if the signal is below the noise level.

The GPS receiver has a single channel and demodulates the signals from each of the four satellites one by one. A receiver determines the range of the satellite using the C/A code or P code.

7.7 CDMA (Code Division Multiple Access)

7.7.1 Orthogonal Chip Sequence

As the name signifies, code division multiplexing is a process of multiplexing where, a number of input channels are multiplexed on the basis of codes, i.e., codes are used as carriers of modulation in the transmitter end [1]. Here, 'code' signifies pseudorandom sequences, chosen properly according to some constraints of orthogonality. In case of spread spectrum modulation, pseudorandom chip sequences are used for bandwidth spreading. Therefore, the frequency of chip sequence should be much higher than the frequency of data stream; i.e., bit duration (T_b) < chip duration (T_c). Here, in case of CDM, direct sequence spread spectrum modulation is used in the transmitter side. But, the constraint is, the chip sequences should be properly defined and orthogonal, so that, no one channel modulated by code sequence 1, can be interfered by other channel modulated by code sequence 2. To satisfy the constraint, the sequences need to follow two distinct properties as follows:

- (i) Cross-correlation Sum of Product is Zero:

If we consider the format of chip and bit sequences as bipolar NRZ, logic 1 is represented by $+1V$ and logic zero is represented by -1 . Now, if we choose

any two sequences of 4 digits, arbitrarily, result of addition of bit by bit product would be 0.

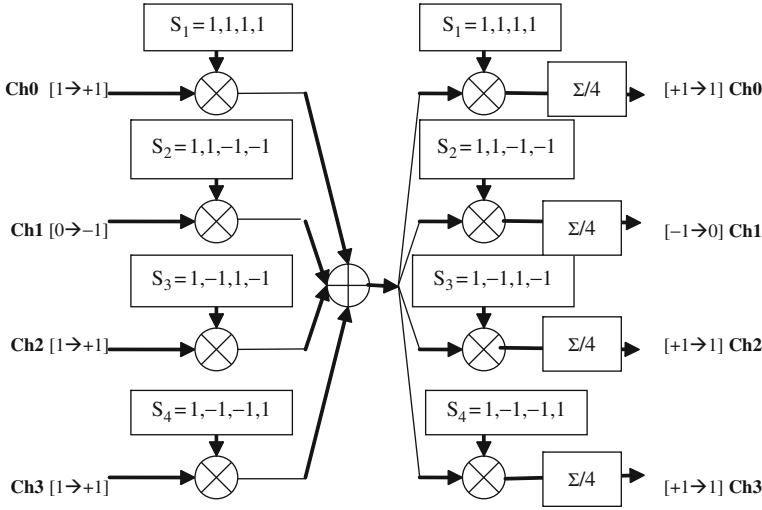


Fig. 7.18 DSSS based CDMA

As for an example (Fig. 7.18), take any two sequences 1010 and 1100.

$$SOP_{cross_bit} = \frac{1}{4} \sum (1, -1, 1, -1) .* (1, 1, -1, -1),$$

where, $.*$ is the bit by bit product (analog multiplication) operator

$$\begin{aligned} \therefore SOP_{cross_Bit} &= \frac{1}{4} \{(1*1) + (-1*1) + (1*-1) + (-1*-1)\} \\ &= \frac{1}{4} \{1 - 1 - 1 + 1\} = 0 \end{aligned}$$

We can test the claim by choosing another pair, say 1001, 1100

$$\begin{aligned} SOP_{cross_Bit} &= \frac{1}{4} \sum (1, -1, -1, 1) .* (1, 1, -1, -1) \\ &= \frac{1}{4} \{(1*1) + (-1*1) + (-1*-1) + (1*-1)\} \\ &= \frac{1}{4} \{1 - 1 + 1 - 1\} = 0 \end{aligned}$$

(ii) Auto-correlation Sum of Product is 'One':

Now, if we choose any sequence of 4 digits ($N=4$), result of addition of bit by bit product would be.

As for an example, take any two sequences 1010.

$$SOP_{auto_bit} = \frac{1}{N} \sum (1, -1, 1, -1) .* (1, -1, 1, -1),$$

where, $.*$ is the bit by bit product (analog multiplication) operator

$$\begin{aligned} \therefore SOP_{auto_Bit} &= \frac{1}{4} \{(1*1) + (-1*-1) + (1*1) + (-1*-1)\} \\ &= \frac{1+1+1+1}{4} = 1 \end{aligned}$$

The chip sequences which follow these two relative constraints are called as orthogonal chip sequences. In our discussion, we have chosen 4 orthogonal chip sequences of 4 digits as, 1111, 1100, 1010, 1001.

7.7.2 Gold Sequence

From the discussion of previous sub-section it is easily understandable that, the number of orthogonal chip sequence increases, as the length of the sequence increases. But, most of the sequences are not suitable for CDMA use. Prof. Kasami and Gold have obtained PN-sequences with better periodic cross-correlation properties. They are derived from the all possible sequences of a particular length and are known as *Gold sequences*. [7, 8]

Gold sequence offer a large number of sequences set with good cross-correlation properties between each pairs. Such sequences are generated by modulo-2 addition of two sequences driven by the same clock as shown in the Fig. 7.19. The most important factor of designing Gold sequence is only special pairs of allowed sequences can give the desired correlation properties.

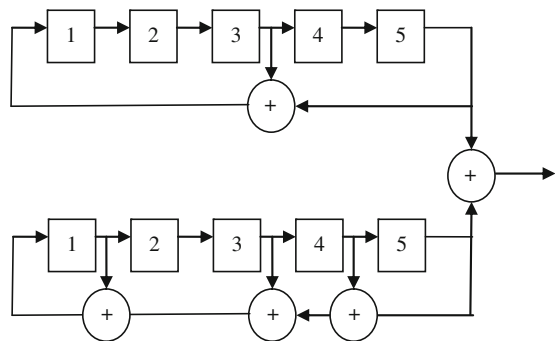


Fig. 7.19 Gold sequence generation

7.7.3 Principle of Operation

7.7.3.1 MUX

According to the figure above (Fig. 7.18), 4 channels are multiplexed into one trunk. Bits sent from Ch0, Ch1, Ch2 and Ch3 are 1, 0, 1, and 1 respectively [1]. These 4 symbols are modulated (DSSS) using seq0 (1111), seq1 (1100), seq2 (1010) and seq3 (1001) respectively, all are in BNRZ format. Therefore, the results of 4 modulators would be

$$\begin{aligned}\text{Ch0_MOD} &\rightarrow (+1, +1, +1, +1) * 1 = (+1, +1, +1, +1) \\ \text{Ch0_MOD} &\rightarrow (+1, +1, -1, -1) * -1 = (-1, -1, +1, +1) \\ \text{Ch0_MOD} &\rightarrow (+1, -1, +1, -1) * 1 = (+1, -1, +1, -1) \\ \text{Ch0_MOD} &\rightarrow (+1, -1, -1, +1) * 1 = (+1, -1, -1, +1)\end{aligned}$$

Adding all the above sequences bit by bit, we get, (2, -2, 2, 2). This is indeed the multiplexed sequence. We can say, this sequence is the multiplexed 1, 0, 1 and 1 after pseudo-code modulation (DSSS). That's why this technique of modulation is called as Code Division Multiplexing, and access to one code division multiplexed line, against permission, is called as CDMA.

7.7.3.2 DMUX

At the de-multiplexer side, firstly, the multiplexed sequence (2, -2, 2, 2) goes to 4 different despreading/demodulator block. The four demodulators are synchronized with proper carrier sequences in order, i.e., seq0 (1111), seq1 (1100), seq2 (1010) and seq3 (1001) respectively.

Therefore, the results of 4 demodulators (analog bit by bit product) would be

$$\begin{aligned}\text{Ch0_DMOD} &\rightarrow (+1, +1, +1, +1) * (2, -2, 2, 2) = (+2, -2, +2, +2) \\ \text{Ch0_DMOD} &\rightarrow (+1, +1, -1, -1) * (2, -2, 2, 2) = (+2, -2, -2, -2) \\ \text{Ch0_DMOD} &\rightarrow (+1, -1, +1, -1) * (2, -2, 2, 2) = (+2, +2, +2, -2) \\ \text{Ch0_DMOD} &\rightarrow (+1, -1, -1, +1) * (2, -2, 2, 2) = (+2, +2, -2, +2)\end{aligned}$$

Next, all the analog values are summed up and normalized to get the transmitted exact bit.

This procedure has a nice advantage that, if even a single digit is erroneous in the carrier chip sequence, the de-multiplexing cannot be done properly, because, spread spectrum; hence the CDMA is highly secure transmission.

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